

# SERKAN YILMAZ

Computer Engineer & Mobile Developer

[GitHub: github.com/serkan-y38](https://github.com/serkan-y38) | [Medium: medium.com/@srkan](https://medium.com/@srkan)

## PROFESSIONAL SUMMARY

Result-oriented and versatile Computer Engineer specializing in building high-performance, scalable mobile applications across Android ecosystems (Flutter, Kotlin). Deeply grounded in Clean Architecture, robust state management, and full-stack integration. Demonstrates a strong technical foundation extending from low-level systems and compiler design to cutting-edge machine learning, IoT, and backend integrations.

## TECHNICAL SKILLS

- **Programming Languages:** Dart, Kotlin, Java, C, C#, Python, SQL
- **Mobile Development:** Flutter, Android SDK, Jetpack Compose, XML, Material 3, Provider, Bloc, Cubit, Push Notifications, Background Tasks, Bluetooth
- **Backend & Cloud Integration:** Spring Boot, Django, Supabase, Firebase, RESTful APIs, MySQL, Database Encryption
- **Architecture & Methodologies:** Clean Architecture, MVVM, Dependency Injection (Dagger Hilt, get\_it), Repository Pattern, OOP Principles
- **Specialized Expertise:** Machine Learning (TensorFlow Lite, ML Kit), Computer Vision (OpenCV), IoT & Hardware (BLE, ESP32), Wearable Technology (Wear OS), Networking (Cisco Packet Tracer)

## SELECTED PROJECTS & PORTFOLIO

### Mobile & Cross-Platform Applications

- **Scalable Full-Stack Flutter App** (*Flutter, Clean Architecture, Bloc, Supabase, RevenueCat*)

- **Augmented Reality (AR) Social App** (*Kotlin, MVVM, AR SceneView, Firebase, Dagger Hilt*)
- **Motivation & Notification App** (*Flutter, Push Notifications, Background Tasks*)
- **Android Bluetooth Chat App** (*Kotlin, Bluetooth API, Jetpack Compose, Coroutines*)
- **Wear OS Interactive Smartwatch Game** (*Kotlin, Wear OS API, Smartwatch UI*)
- **Calm Study App** (*Kotlin, MVVM, Room DB, ML Kit*)

## **Hardware, AI & Computer Vision**

- **ADAS & Vehicle Management Platform** (*OpenCV, TensorFlow Lite, ESP32, BLE, Kotlin*)
- **BLE ESP32 LED Controller** (*C++, Arduino, Android SDK, BLE API, IoT*)
- **Machine Learning & Predictive Modeling Projects** (*Scikit-Learn, Pandas, Regression/Classification*)
- **Computer Vision Processing Suite** (*Python, OpenCV, Object/Facial Detection*)

## **Backend, Low-Level & Core Engineering**

- **SYLMZ Custom Compiler** (*C, Compiler Theory, x86\_64 NASM Assembly, Linux*)
- **Spring Boot Todo RESTful API** (*Java, Spring Boot, MySQL, Authentication*)
- **Django Todo Web Application** (*Python, Django, MVT Architecture*)
- **Network & Routing Infrastructure Projects** (*Subnetting, Cisco Packet Tracer, Network Protocols*)
- **Java Library Management App** (*Java, OOP, Data Structures*)

## **EDUCATION**

**Bachelor of Science in Computer Engineering \* Core Focus: Mobile Development**